

Jet and foil targets for more thrust headroom

V3_30 CAD and its folder masses (14.40 kg). Limits: hover pitch at most 20°, and the rotation from the ground at most 20° (the aircraft only rotates to reach its hover angle). Today's CFD foils (75 / 65 / 50) hover at 43° and rotate 61° from a -17.6° park.

Target: the jets leave each foil group 60° above the fan axis (today's CFD: 39.6°), with the thrust kept as in the CFD (73 N per group). Per group that is **X -36.4 N, Y +63.1 N** (today -56.1 / +46.5). Per station: inner 45°, middle 69°, outer 66° jet angle.

And park level (0°): front foot strut 325 mm (today 100 mm), rear legs 949 mm (today 985). Hover 19.5°, rotation 19.5°.

Flew 3 of 3 in SITL: drift 1.2 m (today's foils 29 m), busiest fan 70 % (today 91 %), no fan saturation, roll authority 4.5 times today's.

Foil angle alone will not get there. In the CFD the jet turns only about 63 % as far as the foil: 75 / 65 / 50 deg of foil gave a 39.6° jet. Across 2,197 foil angle sets the lowest controllable hover was 40°. The CAD change has to keep the jet *attached* through a larger turn (radius, slot, multi-element foil), not just bend the foil further. The foil sets further down are the best possible at today's attachment.

Jet targets that meet both 20° limits

GROUP JET ABOVE FAN AXIS	GROUP FORCE X / Y (N)	STATION JET IN / MID / OUT (DEG)	HOVER DEG	PARK DEG	ROTATION DEG	FRONT STRUT MM	BUSIEST FAN %	SPARE ROLL N M	SITL DRIFT M (WORST)	TOUCH
60° target	-36.4 / +63.1	45 / 69 / 66	19.5	0	19.5	325	70	11.4	1.2 (1.5)	
65°	-30.8 / +66.0	50 / 74 / 71	15.4	-4	19.4	276	67	12.7	1.3 (1.9)	
70°	-24.9 / +68.5	55 / 79 / 76	11.8	-8	19.8	225	64	14.1	4.6 (5.9)	

Thrust kept 67.5 % as in the CFD, 3 x 36 N in per group; CFD frame x along the fan axis, y up. Steeper jets work too (65° and 70° with a -4° / -8° park and a shorter front strut); 60° is the smallest change that meets both limits. SITL: same scenario as below, 3 seeds each, results stamped 2026-10-01T17:28:36-07:00. Ground wobble before lift-off is 4-5° on all three (1.3° for the 35 / 75 / 70 set): a lift-off tuning item, not a design one. Files: `scripts/v3_30_jet_target.py`, `results/v3_30/jet_target/20261001_172609/`.

Parked at -10° (hover 10° or less)

Pick: jets inner 57.5° / middle 85° / outer 85° above the fan axis (alternative 82.5 / 62.5 / 82.5).

Per group **X -17.3 N, Y +68.9 N** (group jet 75.9°). Hover 8.5°, rotation 18.5° from the -10° park, front strut 198 mm, rear legs 972 mm.

Busiest fan 68 %, roll authority 5 times today's; SITL 3 of 3, drift 2.7 m (worst 3.3 m), touchdown 0.16 m/s.

A 10° hover needs the jets almost square to the fan axis: the group must leave about 76° up (today 39.6°). Sweep: 9,261 station triples on 40 to 90° in 2.5° steps, thrust kept 67.5 % and a pessimistic 60 %. 1,258 meet the limits with spare roll and pitch of 5 N m or more and yaw of 1 N m or more, and the busiest fan sits at 67 to 72 % across nearly all of them, so the choice is about the smallest jet turn that still holds the 10° cap when any station is built 2.5° off. The pick and alternative both do; the sets with a 90° station sit on the sweep edge.

JETS IN / MID / OUT (DEG)	GROUP FORCE X / Y (N)	GROUP JET DEG	HOVER DEG	ROTATION DEG	BUSIEST FAN %	SPARE ROLL N M	SPARE YAW N M	SITL DRIFT M (WORST)	TOUCHDOWN M/S	GR BO
57.5 / 85 / 85 pick	-17.3 / +68.9	75.9	8.5	18.5	68.1	12.9	1.35	2.7 (3.3)	0.16	
82.5 / 62.5 / 82.5 alt	-17.6 / +69.7	75.9	8.5	18.5	68.7	10.9	1.52	3.6 (5.7)	0.17	
85 / 65 / 85	-14.5 / +70.4	78.4	7.0	17.1	67.7	11.7	1.53	6.1 (8.1)	0.19	
87.5 / 87.5 / 62.5	-13.3 / +70.1	79.2	6.7	16.7	67.5	10.0	2.48	8.3 (20.0)	0.23	

Results stamped 2026-10-01T17:49:52-07:00. Ground wobble before lift-off grows to 6 to 8° here: a lift-off tuning item. Files: [scripts/v3_30_jet_sweep.py](#), [results/v3_30/jet_sweep/20261001_173459/](#), [airframe](#) [airframes/atlas_v3_30_jets_57.5_85_85_park-10.json](#).

At today's jet attachment: best foil angles

Pick **inner 35° / middle 75° / outer 70°** (alternative 65 / 45 / 75); angle = how far the foil turns the jet, inner = next to the body (M5/M6). Thrust headroom barely moves (busiest fan 92.9 to 91.7 %) because the group force stays about 73 N, but roll authority doubles and SITL drift drops from 29 m to 5 m. It still hovers at 46°, so it does not meet the 20° limits.

Foil sets at today's attachment

FOILS IN / MID / OUT (DEG)	HOVER PITCH	BUSIEST FAN %	SPARE ROLL N M	SPARE PITCH N M	SPARE YAW N M	SITL ROLL ERR DEG	SITL DRIFT M (WORST)	TOUCHDOWN M/S	GROUND BOUNCE DEG
75 / 65 / 50 CFD point	43.3	92.9	2.53	3.41	0.53	0.67	28.6 (62)	0.68	5.6
35 / 75 / 70 pick	46.0	91.7	5.13	3.66	0.70	0.10	5.0 (8)	0.18	1.3
65 / 45 / 75 alt	44.2	91.8	5.71	3.74	0.92	0.18	8.6 (16)	0.21	2.2
45 / 60 / 80	44.0	91.0	5.33	3.13	0.77	0.16	7.2 (13)	0.20	2.6
60 / 50 / 75	43.9	91.5	5.47	3.00	0.75	0.19	8.7 (11)	0.22	2.6

Static columns: the worst of three jet models (below). SITL: Stabilized mode, rotate from the -17.6° park, 0.5 s lift-off ramp, 60 s hover, land; means of 3 seeds, all 15 flights completed (two re-flown after PX4 failed to boot). Spare moment = the most torque on that axis with every fan within 0 to 100 % and the other axes held at hover.

What to check in CFD

Predicted force per foil group (three fans at 36 N each), CFD frame x along the fan axis, y up. Range = across the three jet models. Per-station column: jet angle above the duct axis / thrust kept, for inner, middle, outer (model A).

FOILS	X (N)	Y UP (N)	PER STATION: JET / KEPT
35 / 75 / 70	-59.4 .. -57.0	43.2 .. 43.9	22.2° / 82 % 47.5° / 62 % 44.4° / 65 %
65 / 45 / 75	-58.1 .. -56.8	45.3 .. 45.7	41.2° / 67 % 28.5° / 77 % 47.5° / 62 %
75 / 65 / 50	-56.1 .. -56.1	46.5 .. 46.5	47.5° / 62 % 41.2° / 67 % 31.7° / 75 %

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- Main job: a foil that holds the jet to 60° (per group X -36 N, Y +63 N), force reported per fan.

Fallback run: 35 / 75 / 70. The model's split between stations is the guess here; one group number fits any split.

2 Hold $\pm 2.5^\circ$ on every station.

A 5° error on any one can push the busiest fan to 96 to 98 %, worse than today.

3 Send any second CFD point back.

It decides whether turning harder costs thrust (models A, B) or not (C), which is where the models disagree.

Method

1 Jet model from the CFD point.

Each foil fan: jet angle = $k \times$ foil angle, thrust kept = $1 - \text{loss}(\text{angle})$. Fitted so 75 / 65 / 50 gives the CFD's (-56.12, 46.48) N. $k = 0.634$ (the jet turns about 63 % as far as the foil). Loss: A linear in angle, B quadratic, C flat 32 %.

2 Static sweep.

Every inner / middle / outer triple on 30 to 90° in 5° steps, under all three models: re-trim the hover pitch (PX4 pseudo-inverse mix), then busiest fan share and spare roll / pitch / yaw. Kept triples whose spare moments beat today's foils in every model, ranked by the worst model's busiest fan.

3 SITL.

Each candidate built as a full airframe (trim, MPC_THR_HOVER, nose-lift target, FD_FAIL_P) and flown headless on PX4 1.17, 3 seeds.

Files: `scripts/v3_30_foil_headroom.py` (sweep and airframe build), run folder `results/v3_30/foil_headroom/20261001_165843/` : `calib.json`, `static.json` (6,591 rows, each timestamped), `fly_*.json` and `fly_*_ts/` (flights), `summary.json` / `summary.csv`, airframes `atlas_v3_30_foils_*.json`.

Assumptions: fans 36 N each, all spinning the same way; force points stay on the V3_30 trailing edges; the CFD's 0.73 N spanwise force ignored; only one CFD point, so per-station jet angles are model estimates. Generated 2026-10-01T17:09:17-07:00.